

Heterogeneous temporal graph learning for FX allocation

Research question. Can a heterogeneous temporal graph neural network use macro-financial market blocks to improve currency allocation over a signal-only neural baseline?

The target is not a raw return forecast. The model predicts a long-only allocation vector over an eight-currency basket.

Forex

HTGNN

Mean-variance target

02. Why Forex is a useful financial prediction domain

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The prediction object is relative.

An exchange rate is always defined against another currency. This naturally fits a cross-sectional allocation problem: capital is redistributed across currencies under a common numeraire.

Drivers are macro-financial.

FX is strongly linked to monetary policy, inflation, external balances, global risk appetite, commodities, and rate differentials. These drivers can be grouped into interpretable market blocks.

The project fixes USD as numeraire and allocates across $\mathcal{C} = \{\text{USD, EUR, JPY, GBP, CNY, CAD, AUD, CHF}\}$.

03. Advantages over standard equity-return prediction

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Less firm-idiosyncratic noise

Currencies are less exposed to accounting surprises, management guidance, mergers, litigation, or single-firm shocks.

More accessible context

Rates, commodities, equity-risk proxies, and currency futures provide observable signals that can be organised as typed nodes.

Compact output

Eight currencies produce an interpretable simplex-valued prediction rather than a huge universe of weakly comparable stocks.

This does not make FX easy. It makes the information structure more suitable for a small heterogeneous temporal graph experiment.

No natural productivity-growth premium.

Equities can rise because firms become more productive and expand real cash flows. A currency allocation is mostly relative value under a chosen numeraire.

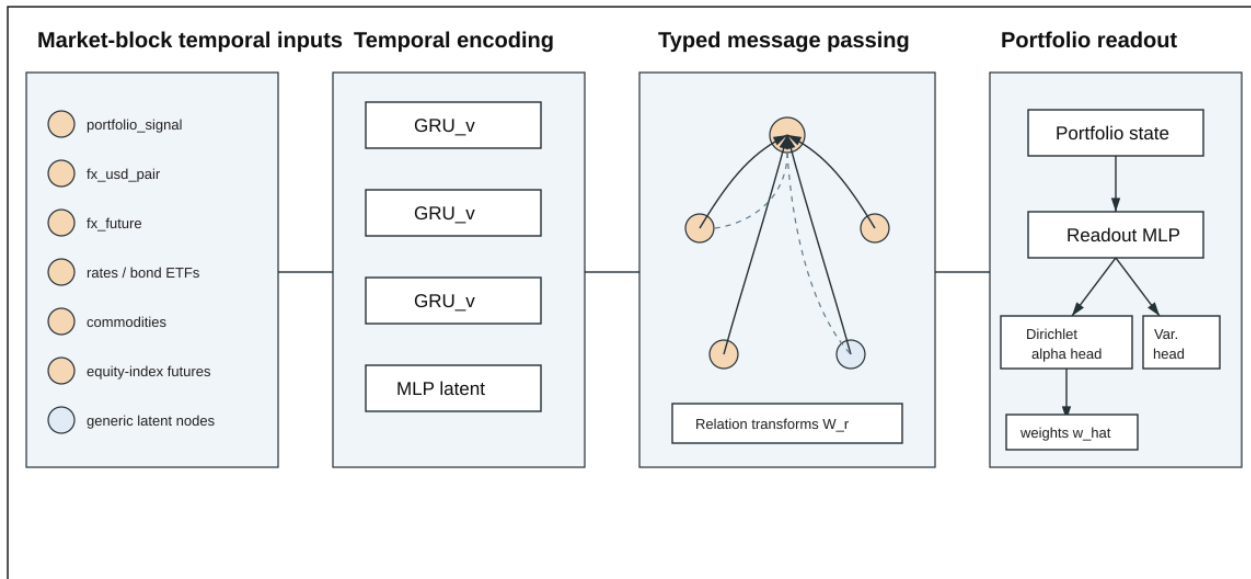
Direct exposure to macro shocks.

Inflation, central-bank intervention, capital controls, geopolitical shocks, and liquidity regimes can dominate short-horizon returns.

The task is therefore useful for testing structured macro-financial modelling, not for claiming that FX returns are easy to predict.

05. Why a heterogeneous temporal GNN

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Raw source. Daily adjusted Yahoo Finance price series.

Period. 2017-01-01 to 2026-01-01.

Transformation. Prices are converted to log returns.

Window. 20 trading days.

Splits. Chronological train, validation, and test splits.

Filtering. Short gaps are forward-filled and highly sparse instruments are removed before tensor construction.

Daily market blocks -> temporal tensors -> supervised allocation dataset

07. Heterogeneous market blocks

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Block	Examples	Forex relevance
commodity_future	GC=F, CL=F, NG=F	Inflation, terms of trade, AUD/CAD sensitivity.
equity_index_future	ES=F, NQ=F, RTY=F	Global risk appetite and safe-haven flows.
us_treasury_future	ZB=F, ZN=F, ZF=F	USD rates curve and duration shocks.
fx_future	6E=F, 6J=F, 6A=F	Liquid alternative FX-market signals.
us_bond_etf	SHY, IEF, TLT, HYG	Rates, credit, inflation, and risk regimes.
fx_usd_pair	USDJPY=X, USDCAD=X	Direct quotes for target construction.

08. USD-denominated currency returns

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The Yahoo Finance quote `USDXXX=X` is interpreted as units of currency `XXX` per one USD. Therefore, the USD-denominated return of holding currency `XXX` uses the opposite sign of the quoted pair return.

$$r_{t, \text{XXX}}^{\text{USD}} = \log \left(\frac{V_t^{\text{XXX}}}{V_{t-1}^{\text{XXX}}} \right) = - \left(\log P_t^{\text{USD/XXX}} - \log P_{t-1}^{\text{USD/XXX}} \right)$$

$$r_{t, \text{USD}}^{\text{USD}} = 0$$

This convention makes all target returns comparable inside the same portfolio simplex.

09. Signal structuring: from returns to allocations

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The project does not ask the network to predict raw future returns directly. Instead, it builds an allocation target from a mean-variance teacher portfolio.

$$\mathbf{w}_t^* = \operatorname{argmax}_{\mathbf{w} \in \Delta_N} \boldsymbol{\mu}_t^T \mathbf{w} - \frac{\lambda}{2} \mathbf{w}^T \boldsymbol{\Sigma}_t \mathbf{w}$$

Input signal. Historical allocation-like sequence derived from past FX returns.

Target signal. Ex-post teacher allocation derived from the future realised cross-section.

10. Why allocation targets are preferable here

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Lower noise objective

The model learns the decision implied by returns and covariance, not every noisy return component independently.

Closer to usage

The final object needed by a portfolio system is w , not an isolated vector of return forecasts.

Less over-exploited

Direct return prediction is a heavily mined target. Allocation imitation encodes a financial decision rule instead.

$$\hat{w}_t = \frac{\alpha_t}{\mathbf{1}^T \alpha_t}, \quad \alpha_t > 0$$

Mean-variance

Uses the model allocation signal or inferred expected returns to trade off return and risk under long-only simplex constraints.

Direct model weights

Uses the predicted simplex vector directly as the portfolio allocation.

Black-Litterman style

Blends prior equilibrium views with model-implied views when uncertainty information is available.

In the current report, the deterministic pipeline is the core analysis target; uncertainty-aware extensions are left for future work.

12. Baseline: deterministic signal-only NN

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The baseline receives only the portfolio signal. It does not receive the heterogeneous market blocks used by the HTGNN.

This is not a fully fair architecture-only comparison, but it is a realistic signal-only financial modelling baseline.

$$\mathbf{z}_t = \varphi_{\theta}(\text{vec}(X_t^{\text{portfolio}}))$$

$$\boldsymbol{\alpha}_t = \text{softplus}(W_{\alpha} \mathbf{z}_t + \mathbf{b}_{\alpha}) + \varepsilon$$

$$\hat{\mathbf{w}}_t = \frac{\boldsymbol{\alpha}_t}{\mathbf{1}^T \boldsymbol{\alpha}_t}$$

13. Results analysis I - evaluation protocol

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Protocol.

- Training period: 2018-01-01 to 2024-01-01
- Validation period: 2024-01-01 to 2025-01-01
- Test period: 2025-01-01 to 2026-01-01
- Daily data, 20-day lookback, same target construction for both models

Reference checkpoints.

- HTGNN-20260507_154855.pt
- NN-20260507_165401.pt

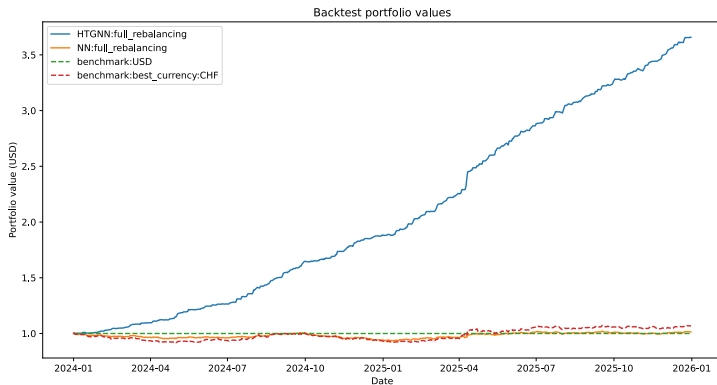
Evaluation focus.

- Direct portfolio performance
- Allocation stability
- Allocation dynamics and interpretability

Model	Train period	Validation period	Test period
NN	2018-2024	2024-2025	2025-2026
HTGNN	2018-2024	2024-2025	2025-2026

14. Results analysis II - supervised allocation metrics

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Main result.

The deterministic HTGNN strongly exceeds all reference comparators over the two-year evaluation window.

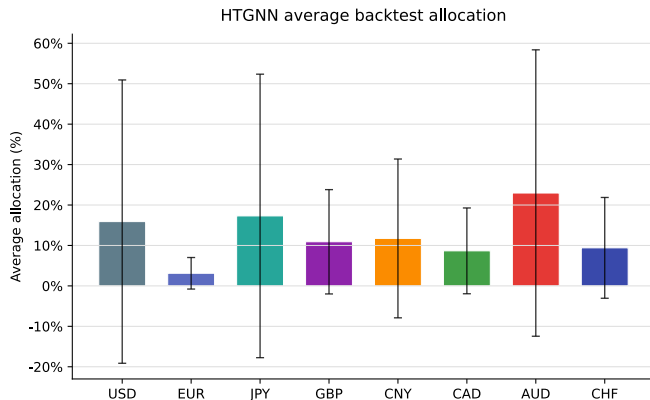
- Cumulative return: **265.7%**
- Annualized return: **87.0%**
- Max drawdown: **-0.64%**

The gap is not marginal; it is a clearly different performance regime.

Reference comparators shown here: deterministic NN, 100% USD, and 100% CHF as the hindsight-best single-currency benchmark.

15. Results analysis III - realised portfolio performance

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Allocation behaviour.

The HTGNN is not just betting permanently on one or two currencies.

Largest average weights:

- AUD: 23.0%
- JPY: 17.3%
- USD: 15.9%

Other meaningful exposures remain active:

- CNY: 11.7%
- GBP: 10.9%
- CHF: 9.4%
- CAD: 8.7%

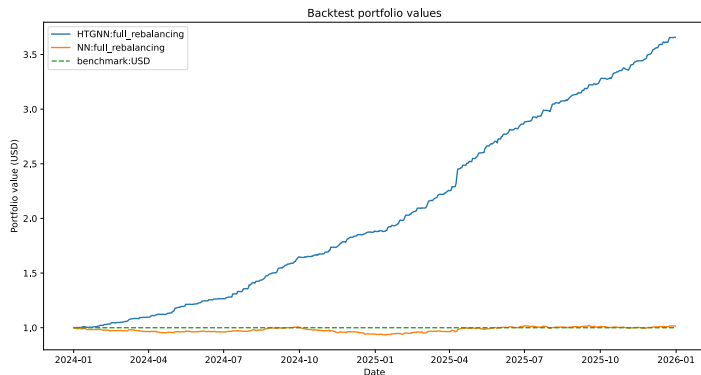
15. Results analysis III - realised portfolio performance (ii)

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So the result is economically more interesting than a degenerate single-currency rule: the portfolio remains relatively balanced while still achieving very high realised performance.

16. Results analysis IV - risk and drawdown diagnostics

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HTGNN vs NN.

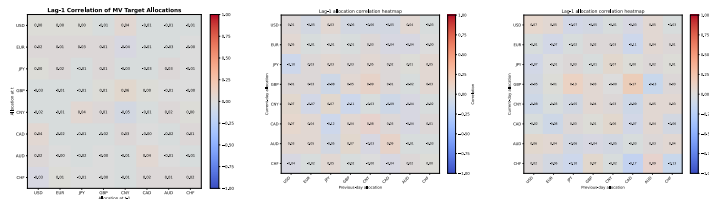
- HTGNN cumulative return: **265.7%**
- NN cumulative return: **1.4%**
- HTGNN annualized return: **87.0%**
- NN annualized return: **0.68%**

The heterogeneous graph clearly dominates the signal-only MLP.

Model	Cum. ret.	Ann. ret.	Sharpe	Max DD
HTGNN	265.7%	87.0%	9.26	-0.64%
NN	1.4%	0.68%	0.14	-7.34%
100% USD	0.0%	0.0%	0.00	0.00%

17. Results analysis V - allocation behaviour

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Lag-1 correlation structure.

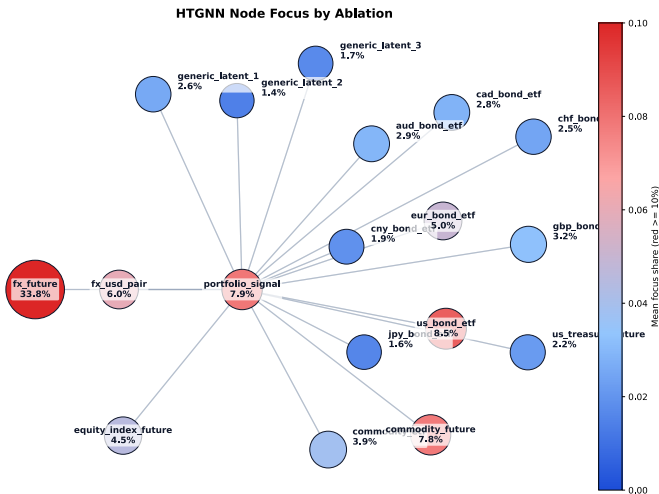
In the input signal and in the HTGNN outputs, lag-1 allocation correlations stay generally low and diffuse.

The NN, in contrast, shows much stronger day-to-day persistence.

Interpretation: this FX task is better suited to combining heterogeneous market blocks than to relying mainly on temporal autocorrelation.

18. Results analysis VI - HTGNN vs NN interpretation

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What the HTGNN uses most.

- fx_future: 33.8%
- us_bond_etf: 8.5%
- portfolio_signal: 7.9%
- commodity_future: 7.8%
- fx_usd_pair: 6.0%

Interesting detail.

Currency-specific treasury proxy nodes receive relatively little attention. The model seems to extract more signal from broad cross-asset macro blocks than from local sovereign-bond proxies alone.

Main takeaways.

- The deterministic HTGNN clearly outperforms the deterministic NN baseline and the market baselines on the two-year evaluation period.
- The gain does not come from simple temporal persistence: unlike the NN, the HTGNN keeps low lag-1 allocation correlations, closer to the structure of the input allocation signal.
- The node-focus analysis is economically plausible: the model relies most on FX futures and broad macro-financial blocks, which supports the value of a heterogeneous graph view of the FX allocation problem.

Bayesian HTGNN extension.

Financial prediction is dominated by uncertainty and partial information. A probabilistic HTGNN could express confidence in allocations and support more conservative decision rules.

Stronger financial evaluation.

Add turnover costs, rolling retraining, stress-period analysis, robustness across seeds, and ablations over market blocks and relation types.

